

Restoration and Segmentation of Retinal Blood Vessels Using CLAHE Enhancement and Morphological Thresholding

Mini Project Report

Digital Image Processing

1. Title

Restoration and Segmentation of Retinal Blood Vessels Using CLAHE Enhancement and Morphological Thresholding.

2. Problem Statement

Retinal fundus images are widely used by ophthalmologists to diagnose vascular and systemic conditions such as diabetic retinopathy, hypertension, and glaucoma. The blood vessel network visible in these images carries important diagnostic information, but manual tracing of vessels by a specialist is time-consuming and subject to human error and fatigue. This project addresses the problem of automatically restoring and segmenting the blood vessel structure from raw retinal fundus images using classical digital image processing techniques, without relying on deep learning or manual annotation at inference time.

The problem has both a restoration component and a segmentation component. Raw fundus images suffer from noise introduced during acquisition and from uneven illumination across the retina, both of which reduce the visibility of thin vessels. Restoration techniques are therefore applied first to clean and enhance the image before segmentation is performed to isolate the vessel structures from the background tissue.

3. Objectives

- To restore retinal fundus images by removing noise and enhancing local contrast so that thin, low-contrast blood vessels become more visible.
- To segment the blood vessel network from the enhanced image using morphological and thresholding-based segmentation techniques.
- To quantitatively evaluate the segmentation output against ground-truth vessel annotations using the Dice coefficient and Intersection over Union (IoU).
- To compare two different thresholding strategies (global Otsu thresholding versus local adaptive thresholding) and identify which performs better for this problem, along with the reasons why.
- To analyze the best-case and worst-case results in order to identify the practical limitations of the chosen approach.

4. Dataset / Image Description

4.1 Source

The DRIVE (Digital Retinal Images for Vessel Extraction) dataset was used for this project. It is a publicly available benchmark dataset created specifically for retinal vessel segmentation research and is commonly mirrored on Kaggle (e.g., [andrewmvd/drive-digital-retinal-images-for-vessel-extraction](https://www.kaggle.com/andrewmvd/drive-digital-retinal-images-for-vessel-extraction)).

4.2 Number of Images and Characteristics

- A subset of 15 images was used from the training split of the DRIVE dataset (image IDs 21 to 35).
- Each image has a resolution of 565 x 584 pixels and is a color (RGB) fundus photograph.
- Each image is accompanied by a ground-truth binary vessel mask (manually annotated by a specialist) and a field-of-view (FOV) mask marking the circular retina region within the square image frame.

4.3 Preprocessing Required

Each image required conversion to a single-channel representation suitable for vessel enhancement (the green channel was selected, as explained in the Methodology section), followed by denoising and contrast enhancement before segmentation could be reliably applied.

4.4 Limitations of the Dataset

- A relatively small subset (15 images) was used, which is suitable for a mini-project but limits the statistical strength of the evaluation.
- Some images contain illumination artifacts (bright glare/reflection regions) which are a known challenge in fundus photography and directly affected segmentation quality, as discussed in the Results section.

5. Methodology

The overall pipeline followed in this project is:

Input Image -> Green Channel Extraction -> Restoration (Median Filtering + CLAHE Contrast Enhancement) -> Segmentation (Black-hat Morphological Filtering + Otsu Thresholding) -> Post-processing (Morphological Opening + Field-of-View Masking) -> Evaluation (Dice / IoU against Ground Truth) -> Result Comparison.

5.1 Green Channel Extraction

Retinal vessels exhibit the highest contrast against the background in the green channel of the RGB image, since the red channel tends to saturate and the blue channel carries very little useful detail. The green channel was therefore extracted and used as the working image for all subsequent steps.

5.2 Restoration: Denoising and Contrast Enhancement

A median filter (kernel size 3) was applied to remove sensor and acquisition noise while preserving the thin edges of the vessels, which a linear filter such as a Gaussian blur would tend to smear. Following denoising, Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied with a clip limit of 2.5 and a tile grid size of 8x8. CLAHE enhances local contrast in small regions of the image independently, which makes faint, low-contrast vessels more visible without over-amplifying noise in flatter background regions, unlike global histogram equalization.

5.3 Segmentation: Black-hat Filtering and Otsu Thresholding

Blood vessels appear as thin, dark, elongated structures against a comparatively brighter retinal background. A morphological black-hat transform (using an elliptical structuring element of size 9x9) was applied to the enhanced image to specifically highlight these thin dark structures while suppressing larger, brighter regions. Otsu's method was then used to automatically compute a global threshold that best separates the highlighted vessel structures from the background, producing a binary vessel mask.

5.4 Post-processing

A morphological opening operation with a small 2x2 kernel was applied to remove isolated noise specks that do not correspond to real vessels. The result was then masked with the dataset's field-of-view mask to restrict the segmentation output to the circular retina region, since the area outside the retina is irrelevant background that should not be evaluated.

5.5 Alternative Method for Comparison: Adaptive Thresholding

To satisfy the requirement of comparing results across different techniques, an alternative segmentation method was also implemented, replacing Otsu's global thresholding with adaptive (locally computed) Gaussian thresholding (block size 15, constant $C = 2$) applied directly to the CLAHE-enhanced image. The same post-processing and evaluation steps were applied to this alternative method so that the two approaches could be compared fairly.

5.6 Evaluation Metrics

Two standard segmentation evaluation metrics were used to compare the predicted vessel mask against the ground-truth mask, restricted to the field-of-view region:

- Dice Coefficient: measures the overlap between the predicted and ground-truth masks, defined as $(2 \times \text{Intersection}) / (\text{Sum of both masks' areas})$.
- Intersection over Union (IoU / Jaccard Index): defined as $\text{Intersection} / \text{Union of the predicted and ground-truth masks}$.

6. Implementation

- Programming Language: Python 3, implemented and executed in Google Colab.
- Libraries Used: OpenCV (image processing operations), NumPy (array operations), Matplotlib (visualization), Pillow/PIL (reading GIF-format ground-truth masks).
- Key Techniques: Green channel extraction, median filtering, CLAHE, morphological black-hat transform, Otsu thresholding, adaptive Gaussian thresholding, morphological opening, field-of-view masking, Dice coefficient, IoU.
- The complete pipeline was implemented as a set of modular functions (preprocess_and_restore, segment_vessels, segment_vessels_adaptive, dice_and_iou) so that each stage could be tested, visualized, and compared independently.

7. Results

7.1 Segmentation Performance (Black-hat + Otsu Thresholding)

The pipeline was run on 15 images from the DRIVE training set. The resulting Dice coefficient and IoU for each image are shown below.

Image ID	Dice Coefficient	IoU
21	0.604	0.433
22	0.595	0.423
23	0.502	0.335
24	0.596	0.425
25	0.530	0.361
26	0.630	0.460
27	0.579	0.407
28	0.605	0.433
29	0.613	0.441
30	0.575	0.404
31	0.574	0.403
32	0.646	0.477
33	0.639	0.470
34	0.468	0.306
35	0.582	0.411

Average Dice Coefficient: 0.583 | Average IoU: 0.413

7.2 Comparison: Otsu Thresholding vs Adaptive Thresholding

The same 15 images were also processed using adaptive thresholding in place of Otsu's global thresholding, keeping every other pipeline stage identical, in order to isolate the effect of the thresholding strategy.

Image ID	Dice (Otsu)	Dice (Adaptive)	IoU (Otsu)	IoU (Adaptive)
21	0.604	0.376	0.433	0.232
22	0.595	0.409	0.423	0.257
23	0.502	0.288	0.335	0.168
24	0.596	0.481	0.425	0.316
25	0.530	0.408	0.361	0.256
26	0.630	0.398	0.460	0.249
27	0.579	0.396	0.407	0.247
28	0.605	0.425	0.433	0.269
29	0.613	0.400	0.441	0.250
30	0.575	0.380	0.404	0.234
31	0.574	0.311	0.403	0.184
32	0.646	0.413	0.477	0.260
33	0.639	0.393	0.470	0.245
34	0.468	0.376	0.306	0.232
35	0.582	0.393	0.411	0.244

Average Dice - Otsu: 0.583 | Adaptive: 0.390 Average IoU - Otsu: 0.413 | Adaptive: 0.243

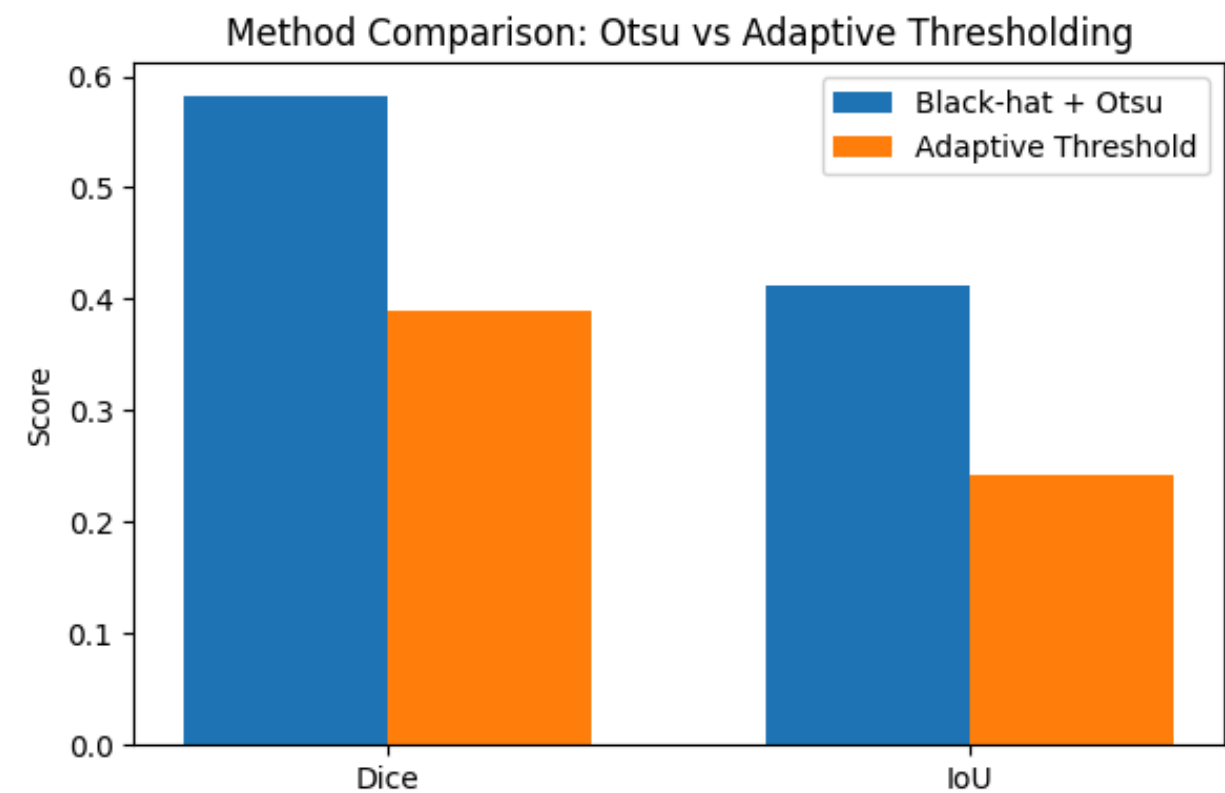


Figure 1: Average Dice and IoU comparison between Black-hat + Otsu thresholding and Adaptive thresholding.

Otsu's global thresholding consistently outperformed adaptive thresholding across every one of the 15 test images, for both Dice and IoU. This indicates that the locally-computed adaptive threshold was more sensitive to small-scale brightness and texture variation in the background

tissue, causing it to misclassify background texture as vessel pixels (false positives), which reduced both the Dice score and the IoU.

7.3 Best Case and Worst Case Analysis

To understand the boundaries of the method's performance, the best-scoring and worst-scoring images from the Otsu-based pipeline were examined in detail.

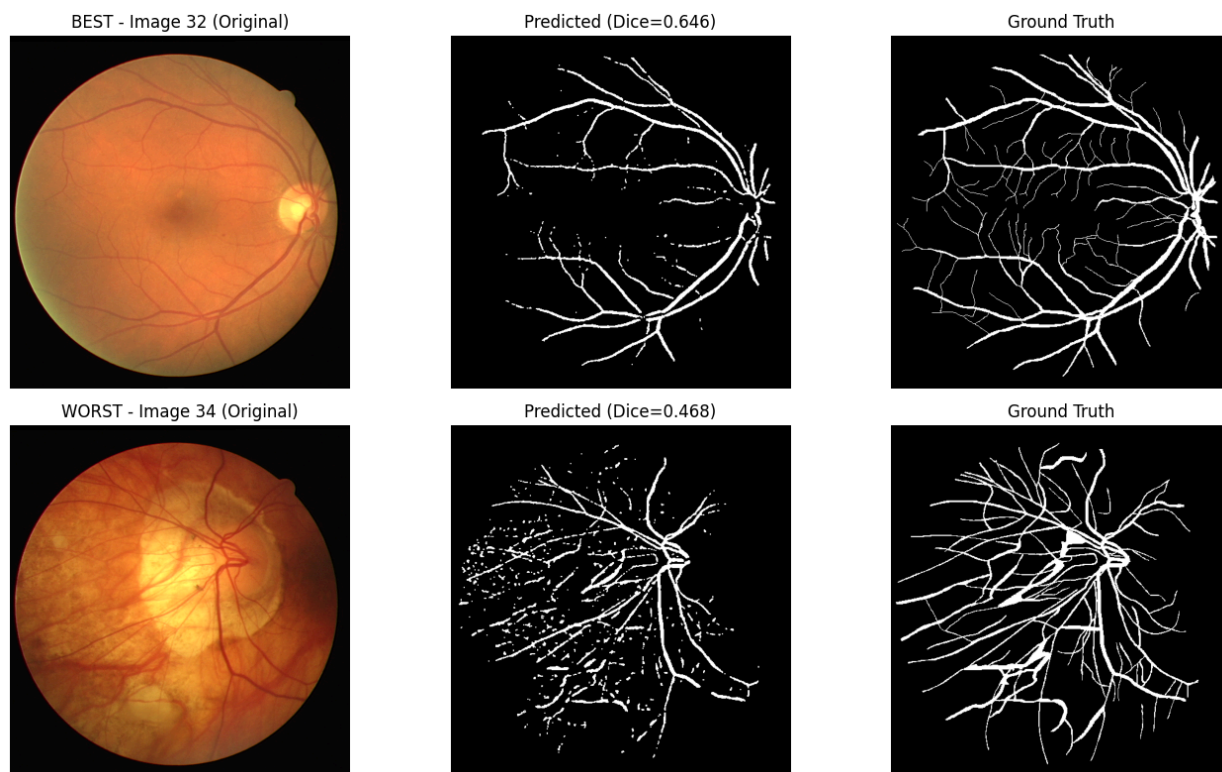


Figure 2: Best-case (Image 32, Dice = 0.646) versus worst-case (Image 34, Dice = 0.468) segmentation results, compared against their respective ground-truth vessel masks.

Best Case (Image 32, Dice = 0.646): This image has relatively uniform background illumination and good contrast between the vessels and the surrounding tissue. The predicted mask captures the major vessel branches accurately, though some of the thinnest peripheral vessels visible in the ground truth were missed.

Worst Case (Image 34, Dice = 0.468): This image contains a large bright glare or reflection artifact on one side, along with generally uneven illumination. The black-hat filter mistook the edges of this bright artifact for vessel-like dark structures, producing scattered false-positive speckles concentrated in that region. At the same time, genuine thin vessels in the image were harder to detect due to lower local contrast, resulting in missed detections.

8. Limitations

- Small image subset: only 15 images from the DRIVE dataset were used, which is sufficient to demonstrate the method but too small to draw strong statistical conclusions.

- Sensitivity to illumination artifacts: images containing glare, reflections, or strongly uneven brightness (as in the worst-case image) produce noticeably lower segmentation accuracy.
- Missed thin vessels: the thinnest, faintest peripheral vessels are frequently missed in both the Otsu and adaptive thresholding approaches, since their contrast against the background is close to the noise floor.
- Sensitivity to parameter tuning: results depend on manually chosen parameters (CLAHE clip limit, black-hat structuring element size, adaptive threshold block size), which were tuned by inspection rather than through a systematic search.
- No deep-learning baseline: the project intentionally uses only classical DIP techniques as required by the assignment; state-of-the-art vessel segmentation using deep learning was not evaluated for comparison.

9. Future Scope

- Evaluate the pipeline on the full DRIVE dataset (40 images) and on additional retinal datasets (e.g., STARE, CHASE_DB1) to test generalization.
- Apply illumination correction (e.g., homomorphic filtering or background subtraction) prior to segmentation to reduce sensitivity to glare artifacts such as the one observed in the worst-case image.
- Explore multi-scale vessel enhancement filters (e.g., Frangi vesselness filter) to better capture both thick and thin vessels simultaneously.
- Compare against a deep-learning-based segmentation model (e.g., U-Net) as a stronger baseline for accuracy, while retaining the classical pipeline as a lightweight, training-free alternative.
- Package the pipeline into a simple web or desktop tool for real-time vessel visualization, which could support screening applications.

10. Conclusion

This project developed a classical digital image processing pipeline to restore and segment blood vessels from retinal fundus images using the DRIVE dataset. Green channel extraction, median filtering, and CLAHE contrast enhancement were used to restore image quality, while black-hat morphological filtering combined with Otsu thresholding was used to segment the vessel network. The approach achieved an average Dice coefficient of 0.583 and an average IoU of 0.413 across 15 test images. A comparison against adaptive thresholding showed that global Otsu thresholding was consistently more accurate for this problem, and a best-case/worst-case analysis revealed that illumination artifacts such as glare are the primary cause of degraded performance. While the method does not match the accuracy of modern deep-learning approaches, it demonstrates that simple, well-understood, and computationally inexpensive classical techniques can meaningfully restore and segment medically relevant structures in retinal images, and it establishes a clear, explainable baseline together with concrete directions for future improvement.